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Course: Calculus Beta

Assignment: MyLab Math - Uge 4

Find the three cube roots of $-64i$.

If $z = re^{i\theta}$, then the expression $\sqrt[n]{re^{i\theta}}$ refers to the following collection of n distinct numbers. Note that $\sqrt[n]{r}$ refers to the single real n th root of r .

$$\sqrt[n]{re^{i\theta}} = \sqrt[n]{r} \exp \left[i \left(\frac{\theta}{n} + k \frac{2\pi}{n} \right) \right], k = 0, 1, 2, \dots, n-1$$

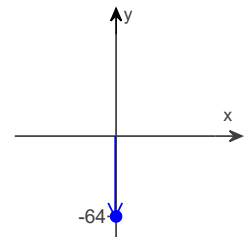
Since the goal in this case is to find the cube roots of a number, $n = 3$.

The expression for $\sqrt[n]{re^{i\theta}}$ involves $\frac{2\pi}{n}$. Since $n = 3$, $\frac{2\pi}{n} = \frac{2\pi}{3}$.

This means the cube roots of $z = re^{i\theta}$, w_0 , w_1 , and w_2 , are given by the following formula.

$$w_k = \sqrt[3]{r} \exp \left[i \left(\frac{\theta}{3} + k \frac{2\pi}{3} \right) \right], k = 0, 1, 2$$

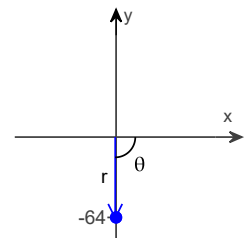
Plot $-64i$ in the complex plane and determine its polar representation $re^{i\theta}$. This figure shows $-64i$ in the complex plane.



Use this figure to find the values of r and θ , with $r \geq 0$ and $-\pi < \theta \leq \pi$, for the number $-64i$.

$$r = 64$$

$$\theta = -\frac{\pi}{2}$$



$$\text{This means } -64i = 64e^{-\frac{i\pi}{2}}.$$

Substitute $r = 64$ and $\theta = -\frac{\pi}{2}$ into the formula for the cube roots of a number from earlier.

$$\begin{aligned} w_k &= \sqrt[3]{r} \exp \left[i \left(\frac{\theta}{3} + k \frac{2\pi}{3} \right) \right] \\ &= \sqrt[3]{64} \exp \left[i \left(\frac{-\frac{\pi}{2}}{3} + k \frac{2\pi}{3} \right) \right] && \text{Substitute } r = 64 \text{ and } \theta = -\frac{\pi}{2}. \\ &= 4 \exp \left[i \left(-\frac{\pi}{6} + k \frac{2\pi}{3} \right) \right] && \text{Simplify.} \end{aligned}$$

Let $k = 0$ in the formula for w_k to find an expression for one cube root, w_0 .

$$\begin{aligned} w_k &= 4 \exp \left[i \left(-\frac{\pi}{6} + k \frac{2\pi}{3} \right) \right] \\ w_0 &= 4 \exp \left[i \left(-\frac{\pi}{6} + (0) \frac{2\pi}{3} \right) \right] && \text{Substitute } k = 0. \\ &= 4e^{-\frac{i\pi}{6}} && \text{Simplify.} \end{aligned}$$

Euler's Formula states that $e^{i\theta} = \cos \theta + i \sin \theta$. Use this to rewrite the expression from the previous step in terms of sine and cosine.

$$w_0 = 4e^{-\frac{i\pi}{6}}$$

$$= 4 \left[\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right] \quad \text{Use Euler's Formula.}$$

Evaluate $\cos \left(-\frac{\pi}{6} \right)$ and $\sin \left(-\frac{\pi}{6} \right)$.

$$w_0 = 4 \left[\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right]$$

$$= 4 \left[\frac{\sqrt{3}}{2} + i \left(-\frac{1}{2} \right) \right] \quad \text{Evaluate.}$$

This means one cube root of $-64i$ is $w_0 = 4 \left[\frac{\sqrt{3}}{2} - \frac{1}{2}i \right] = 2\sqrt{3} - 2i$.

Let $k = 1$ in the formula for w_k to find an expression for the second cube root, w_1 .

$$w_k = 4 \exp \left[i \left(-\frac{\pi}{6} + k \frac{2\pi}{3} \right) \right]$$

$$w_1 = 4 \exp \left[i \left(-\frac{\pi}{6} + (1) \frac{2\pi}{3} \right) \right] \quad \text{Substitute } k = 1.$$

$$= 4e^{\frac{i\pi}{2}} \quad \text{Simplify.}$$

Use Euler's Formula to rewrite this in terms of sine and cosine.

$$w_1 = 4e^{\frac{i\pi}{2}}$$

$$= 4 \left[\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right] \quad \text{Use Euler's Formula.}$$

Evaluate $\cos \frac{\pi}{2}$ and $\sin \frac{\pi}{2}$.

$$w_1 = 4 \left[\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right]$$

$$= 4[0 + i(1)] \quad \text{Evaluate.}$$

This means the second cube root of $-64i$ is $w_1 = 4[0 + i] = 4i$.

Let $k = 2$ in the formula for w_k to find an expression for the third cube root, w_2 .

$$w_k = 4 \exp \left[i \left(-\frac{\pi}{6} + k \frac{2\pi}{3} \right) \right]$$

$$w_2 = 4 \exp \left[i \left(-\frac{\pi}{6} + (2) \frac{2\pi}{3} \right) \right] \quad \text{Substitute } k = 2.$$

$$= 4e^{\frac{7i\pi}{6}} \quad \text{Simplify.}$$

Use Euler's Formula to rewrite this in terms of sine and cosine.

$$w_2 = 4e^{\frac{7i\pi}{6}}$$

$$= 4 \left[\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right] \quad \text{Use Euler's Formula.}$$

Evaluate $\cos \frac{7\pi}{6}$ and $\sin \frac{7\pi}{6}$.

$$w_2 = 4 \left[\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right]$$

$$= 4 \left[-\frac{\sqrt{3}}{2} + i \left(-\frac{1}{2} \right) \right] \quad \text{Evaluate.}$$

This means the third cube root of $-64i$ is $w_2 = 4 \left[-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right] = -2\sqrt{3} - 2i$.

Thus, the three cube roots of $-64i$ are $2\sqrt{3} - 2i$, $4i$, and $-2\sqrt{3} - 2i$.